ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

MECHANICAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Mechanical Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers.

QUESTION 1

1 1.1 C ✓✓ 1.2 A ✓✓ 1.3 A ✓✓ 1.4 B ✓✓ 1.5 B ✓✓ [10]

QUESTION 2

2.1 Any FOUR: wear overalls and safety shoes ✓; wear safety glasses ✓; no loose clothing, jewellery or long loose hair near machines ✓; never leave a running machine unattended ✓; use guards; report injuries; keep walkways clear (4)

2.2 Any THREE: wear eye protection / use the eye shield ✓; keep the tool rest 2–3 mm from the wheel ✓; stand to the side when starting it ✓; check the wheel for cracks; never grind on the side of the wheel; keep guards in place (3)

2.3 Any TWO dangers with a precaution, 2 marks each: fire or explosion from leaking gas ✓ → check hoses and use flashback arrestors / soapy water leak test ✓; burns from hot metal ✓ → wear leather gloves and apron ✓; eye damage from the bright light ✓ → wear welding goggles ✓; toxic fumes → work in a ventilated area (4)

2.4 Carbon dioxide (CO_2) ✓✓ (accept dry chemical powder); never use water (2)

2.5 People can slip and fall and be injured ✓; oil is also a fire hazard ✓. (2)

[15]

QUESTION 3

3.1 Centre punch: make a small dent so a drill starts in the right place ✓; try square: check and mark right angles ✓; vernier caliper: measure inside, outside and depth dimensions accurately ✓; micrometer: measure very small dimensions very accurately ✓; hacksaw: cut metal ✓ (5)

3.2 $7 \times 0{,}1 = 0{,}7 \text{ mm}$ ✓; $24 + 0{,}7$ ✓ $= 24{,}7 \text{ mm}$ ✓ (3)

3.3 The teeth must point forward (away from the handle) ✓ and the blade must be tight ✓. For thin sheet metal, use a fine blade (many teeth per 25 mm, e.g. 32 TPI) so that at least three teeth touch the metal ✓. (3)

3.4 Any THREE: flat ✓; round ✓; half-round ✓; square; triangular (three-square); hand file (3)

3.5 Any THREE: clean them after use ✓; store them in their cases ✓; do not drop them ✓; do not over-tighten micrometers; protect from rust with a light oil; check their accuracy (zero) regularly (3)

3.6 $$\text{N} = \frac{30}{3{,}14 \times 0{,}01}$$ ✓✓ $= 955{,}41 \text{ r/min}$ ✓ (3)

[20]

QUESTION 4

4.1 Ferrous metals contain iron ✓, e.g. mild steel, cast iron ✓. Non-ferrous metals contain no iron ✓, e.g. copper, aluminium, brass ✓. (4)

4.2 Hardness: the ability to resist scratching, wear and indentation ✓✓. Toughness: the ability to absorb energy (resist shock or impact) without breaking ✓✓. Malleability: the ability to be hammered or rolled into thin sheets without cracking ✓✓. (6)

4.3 (a) Hardening makes steel harder and more wear-resistant ✓✓ (heat, then quench). (b) Annealing softens metal and relieves internal stresses, making it easier to work ✓✓ (heat, then cool slowly). (4)

4.4 Brass: copper ✓ and zinc ✓. Bronze: copper ✓ and tin ✓. (4)

4.5 Any TWO: light in weight ✓; strong for its weight ✓; resists corrosion (2)

[20]

QUESTION 5

5.1 Permanent joints cannot be taken apart without damaging the parts ✓, e.g. welding, riveting ✓. Temporary joints can be taken apart and put together again ✓, e.g. nuts and bolts, screws ✓. (4)

5.2 Any THREE: welding helmet with a dark filter lens ✓; leather gloves ✓; leather apron / jacket ✓; safety boots; spats (3)

5.3 Any ONE with a reason: joining aluminium sheets in aircraft or trailers, because heat from welding weakens the metal ✓✓ / joining different metals / where welding is not possible on site. (2)

5.4 It spreads the load over a bigger area ✓ and protects the surface from damage (spring washers stop the nut from loosening) ✓. (2)

5.5 In soldering, only a filler metal with a low melting point is melted ✓; the parent metal does not melt ✓. In welding, the parent metals themselves are melted and fused together ✓, giving a much stronger joint. (3)

5.6 Pitch: the distance from a point on one thread to the same point on the next thread ✓. Crest: the top (outermost) surface of the thread ✓. Root: the bottom of the groove between two threads ✓. (3)

5.7 $30 \div 1{,}5$ ✓✓ $= 20$ threads ✓ (3)

[20]

QUESTION 6

6.1 Regular, planned servicing of machines before they break down ✓✓. Benefits, any TWO: fewer breakdowns ✓; longer life of machines ✓; safer working; lower repair costs (4)

6.2 Any TWO: reduces friction and wear ✓; cools moving parts ✓; prevents rust; removes dirt (2)

6.3 $\text{Moment} = \text{F} \times \text{d} = 80 \times 0{,}25$ ✓✓ $= 20\ \text{N·m}$ ✓ (3)

6.4 Tension: a towing cable / crane rope / bolt being tightened ✓. Compression: a car jack / pillar / connecting rod on the power stroke ✓. Shear: a rivet or bolt joining two plates / scissors cutting ✓. (3)

6.5 $$N_{driven} = \frac{N_{driver} \times D_{driver}}{D_{driven}} = \frac{1\,400 \times 100}{200} \checkmark\checkmark = 700 \text{ r/min} \checkmark$$ (3)

[15]

TOTAL: 100

END OF MARKING GUIDELINE